



Society of Petroleum Engineers

SPE-229133-MS

Technical Pathways for Middle East Data Center Decarbonisation: Leveraging Global Best Practices and GCC Case Studies

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229133-MS](https://doi.org/10.2118/229133-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

This manuscript aims to analyse technical, regulatory, and economic pathways for GCC data center decarbonization, informed by global practices. While data centers offer significant monetary benefits through revenue generation and investment, their power consumption is a critical factor that needs to be managed. The balance between economic gains and energy efficiency is essential for sustainable growth in the data center industry. The objectives of this paper include evaluating decarbonization solutions, reviewing regulatory frameworks, assessing historical and future energy demand, proposing a decarbonization index, and examining technical and economic viability. The scope also covers established and emerging technologies applicable globally.

This study employs a multidisciplinary approach combining technical evaluation, regulatory benchmarking, and economic modelling. Energy demand trends, both historical and projected, are synthesized to establish baseline and future scenarios. Technical assessments focus on efficiency-enhancing strategies such as liquid cooling, waste heat reuse, and AI-driven load management. Decarbonization technologies including carbon capture and storage (CCS), direct air capture (DAC), and various renewables, energy storage solutions are evaluated based on feasibility, scalability, and cost-effectiveness. Deployment models are analysed across onshore, offshore, and repurposed infrastructures. External decarbonization contributions, such as integration with renewable-powered grids and provision of grid-balancing services, are also examined. A quantifiable decarbonization index is developed to benchmark sustainability progress. Financial modelling compares conventional and decarbonized data centre scenarios under GCC-specific economic and policy conditions to assess profitability and bankability.

Data centre energy demand is projected to rise, intensifying the need for effective decarbonization. Regulatory frameworks, both global and regional, are increasingly pushing for low-carbon transitions. While advanced technologies offer deep decarbonization potential, they remain constrained by high costs and deployment complexity. Energy storage systems facilitate renewable integration but require strategic investment. Offshore asset repurposing emerges as a promising pathway, offering cooling benefits, efficiency, and cost savings from avoided decommissioning. The proposed decarbonization index

provides a standardized metric for comparing sustainability performance. Economic modelling indicates that, under supportive policy frameworks or revenue from ancillary grid services, decarbonised data centers can achieve financial viability. Moreover, data centers can contribute meaningfully to broader energy system decarbonization, especially when designed as flexible, grid-interactive assets. Regional case studies underscore the importance of context-specific strategies blending efficiency, renewable energy, and emerging technologies to achieve sustainable operations.

This paper offers a novel, integrated approach and framework for GCC data center decarbonisation, addressing a critical global challenge and providing the best set of alternatives through specific regional case studies. Key additions include: a proposed quantifiable decarbonisation index; specific technical-economic analysis of offshore asset repurposing linked to decommissioning; and comparative profitability modelling for the GCC. This approach can easily be replicable globally, considering the local regulations and constraints.

Introduction

Data Centers form the foundation of the digital economy, yet their energy intensive consumption poses significant environmental challenges, necessitating urgent decarbonization. Housing a complex array of servers, storage, and networking equipment, supported by critical power and cooling infrastructure, these facilities have a substantial carbon footprint. This paper examines the imperative and pathways for data center decarbonization within the Gulf Cooperation Council (GCC) region, where rapidly growing digital demand aligns with global trends.

A data center (DC) is a facility used to house computer systems and associated components, such as telecommunications and storage systems with important core elements such as servers, storage systems, networking equipment, power supply, cooling systems, and security (Smith 2023a). Their functions are related to data processing, handling large volumes of data, storage, networking, and disaster recovery, ensuring data integrity (Brown 2022). There are different types of data centers: 1) enterprise DC (company's needs), 2) colocation DC (multiple organizations), 3) Cloud DC (operated by cloud service provider), and 4) edge DC (smaller facility closer to end-users) (Johnson 2021).

With a single AI query potentially requiring 10x the electricity of a typical internet search, this growth will have major implications for the global power market with demand set to increase by 160% by 2030 (Goldman Sachs 2024). On the other hand, the power & utility sector is one of the most significant contributors to global greenhouse gas emissions (GHG), responsible for approximately 35% of total emissions. This means a significant increase in emissions and highlights the importance of focusing on decarbonizing data centers (Global Data 2025).

This manuscript analyses the historical context, global and regional drivers, evolving regulatory landscapes, and critically, the technical solutions applicable to the GCC's unique operational environment, including climate considerations impacting cooling demand and opportunities presented by regional renewable energy resources and economic diversification strategies. The study explores viable technical interventions across various data center types and evaluates the expected outcomes, providing a focused perspective on achieving sustainable digital infrastructure essential for the GCC's continued technological advancement.

Historical Background

The evolution of data centers is a journey that mirrors the advancements in computing technology and the growing demand for data processing and storage. The following table shows the data center concept evolution as described in the below literature.

Table 1—Evolution of Data Centers

Timeline	Features Evolution
1960s: Early Computer Rooms	Mainframes: The concept of data centers began with mainframe computers housed in dedicated rooms. These early data centers were highly controlled environments needed for the massive, room-filling computers of the time (Miller, et al. 2002).
1970s: Emergence of the Term ‘Data Center’	Specialized Rooms: As computing power grew and became central to business operations, the term "data center" started to gain use. More organizations, especially financial institutions and governments, began to build specialized rooms for their computer equipment (Miller, et al. 2002).
1980s: Rise of Client-Server Architecture	Expansion: The transition from mainframes to client-server architecture increased the need for multiple servers and storage devices, leading to more expansive data centers. Growth in telecommunications also contributed to an increase in the number of data centers (Miller, et al. 2002).
1990s: Dot-Com Boom and Internet Expansion	Proliferation: With the internet's rapid growth and the dot-com boom, data centers proliferated. They expanded in size and number to support web hosting, e-commerce, and the burgeoning needs of internet-based businesses. Colocation data centers also emerged, where multiple companies shared space in a single facility (Miller, et al. 2002). In 1998, data centers were relatively small and consumed significantly less energy compared to today. The footprint of data centers in terms of megawatts consumption was minimal, reflecting the nascent stage of digital infrastructure.
2000s: Advent of Cloud Computing and Virtualization	Large-Scale Facilities: Cloud computing and virtualization technologies revolutionized the data center industry. Large-scale data centers were built to support the services provided by tech giants like Google, Amazon, and Microsoft. Facilities became more energy-efficient, and hyperscale data centers (over 5,000 servers) started to appear. In 2006, Google launched the first hyperscale data center in The Dalles, Oregon with a hyperscale facility which occupied 1.3 million square feet of space and employs a staff of approximately 200 data center operators. (IBM 2025)
2010s: Hyperscale and Edge Data Centers	Massive Growth: The rapid expansion of cloud services, big data, and the Internet of Things (IoT) has significantly increased the demand for larger data centers. Hyperscale data centers have become the standard, while edge data centers bring data closer to users, improving speed and reducing latency (Greenstein & Fang 2022; BCG 2025).
2020s: Continued Expansion and Diversification	Modern Era: The growth continues with increasing demand for cloud services, AI, 5G, and edge computing. The largest data centers now exceed 1 million square feet, and there are over 600,000 data centers globally (Harvard GSD 2024). Regulations: Over time, regulations have evolved to include environmental considerations, energy efficiency standards, and incentives for renewable energy adoption (Harvard GSD 2024; Edge Computing Lab 2024).

In recent years, data centers have become increasingly innovative to meet the growing demands of modern technology. One of the most notable examples is the Tahoe Reno data center in Nevada, US. This facility, located in the Tahoe Reno Industrial Center, is one of the largest colocation facilities globally, offering 7.2 million square feet of data center space (Srivastava 2024). Another remarkable example is the State Line Coal Plant in Indiana, US, which has been transformed from a former coal-fired power plant into a green data center. This facility uses water and wind from Lake Michigan for cooling and repurposes excess heat to warm greenhouses, showcasing a creative approach to sustainability (Srivastava 2024).

Another innovative data center is the Nautilus Data Center in Limerick, Ireland, which is a floating data center that leverages water cooling and cheaper real estate. This unique approach aims to utilize ports with spare capacity worldwide (Srivastava, 2024). Additionally, the Green Mountain data center in Norway, located in a former NATO ammunition storage facility, uses 100% renewable hydropower and fjord water for cooling. It is designed to withstand electromagnetic pulses and nuclear blasts, highlighting its resilience and sustainability (Srivastava 2024). These examples illustrate the diverse and complex landscape of modern data centers, which are continually evolving to address environmental concerns, enhance efficiency, and meet the specific needs of various industries.

Existing Data Center Index

Power Usage Effectiveness (PUE) is a crucial metric for assessing the energy efficiency of data centers. PUE is defined as the ratio of the total amount of energy consumed by a data center facility to the energy consumed by the IT equipment within that facility. This metric helps determine how efficiently a data center uses energy, specifically highlighting the proportion of energy consumed by computing equipment compared to the overall energy consumption, which includes cooling, lighting, and other overheads (The Green Grid

2007). A lower PUE value indicates a more energy-efficient data center, as it signifies that a larger portion of the energy is being used directly for computing purposes rather than for ancillary functions.

Understanding and optimizing PUE is essential for data center operators aiming to reduce operational costs and minimize environmental impact. For instance, advanced cooling technologies, such as liquid cooling and free cooling, can significantly improve PUE by reducing the energy required for temperature regulation (Data Center Knowledge 2023). Additionally, implementing energy-efficient IT equipment and optimizing data center layout can significantly enhance energy efficiency. The ideal PUE is 1.0, indicating that all the energy consumed by the data center is used by the IT equipment, though achieving this is practically challenging (Data Center Knowledge 2023). Continuous monitoring and improvement of PUE are essential for achieving sustainable and cost-effective data center operations. Advanced cooling technologies, such as liquid cooling and free cooling, can help reduce the energy required for temperature regulation, thereby improving PUE (PUE Calculator 2023).

The formula for calculating PUE is:

$$PUE = \frac{\text{Total Facility Energy}}{\text{IT Equipment Energy}}$$

Total Facility Energy: This includes all the energy consumed by the data center, such as cooling systems, lighting, power conversion, uninterruptible power supplies (UPS), and other non-IT equipment.

IT Equipment Energy: This is the energy consumed by the core IT equipment, including servers, storage devices, and networking equipment.

Other factors that could affect PUE are climate, operational load, facility size, cooling technologies, and energy sources. PUE does not account for external climate conditions, so data centers in colder climates may show lower PUE despite having similar efficiency to those in warmer regions. It is also most meaningful when IT equipment operates at full capacity, as load variations can affect energy efficiency measurements. Larger data centers tend to be more efficient due to economies of scale and newer technologies (The Green Grid 2021). Innovations such as liquid cooling and free cooling help reduce energy use (Vertiv 2023). The use of renewable energy also contributes to improved efficiency and reduced environmental impact. Additionally, operational load plays a role, with higher IT utilization typically leading to better energy efficiency.

In the late 2010s, the average PUE continued to improve, but at a slower rate, stabilizing around 1.58 by 2019. This period saw the adoption of more advanced cooling technologies and energy-efficient IT equipment, contributing to incremental gains in efficiency (Mobile Data Centre 2021). In recent years, the average annual PUE for global data centers has remained relatively stable, ranging from 1.55 to 1.59. Despite ongoing modernization efforts, the overall PUE figure has not changed dramatically due to the presence of older, less-efficient facilities that continue to operate (The Green Grid 2021). Continuous monitoring and improvement of PUE remain essential for achieving sustainable and cost-effective data center operations.

Carbon Usage Effectiveness (CUE) is also a commonly referenced metric for understanding the carbon footprint of data centers, but it provides only a partial view of overall carbon performance.

The formula for calculating CUE is:

$$CUE = \frac{\text{Total CO}_2 \text{ Emissions from Energy Consumption}}{\text{IT Equipment Energy}}$$

Total CO₂ Emissions: Calculated using the facility's total energy consumption and the emissions factor of its energy source (e.g., grid mix).

IT Equipment Energy: Energy consumed by core computing infrastructure such as servers, storage, and networking.

In theory, a lower CUE suggests that a data center is powered by cleaner energy sources or is otherwise emitting less CO₂ per unit of IT energy. However, CUE has key limitations:

- It does not account for embodied emissions in hardware or infrastructure.
- It ignores facility-level energy use outside of IT equipment, which may be substantial in older or poorly optimized sites.
- It treats all IT energy as equally productive, failing to differentiate between high-performance computers (e.g., token generation in AI models) and idle or low-value workloads (Soni et al. 2025).

While CUE can be helpful as a directional indicator, particularly in benchmarking energy procurement strategies, it should not be viewed as a comprehensive measure of carbon efficiency. For instance, two data centers with identical CUE scores might have vastly different digital output per unit of carbon emitted, depending on workload type, energy utilization, and system efficiency and is insufficient for capturing the full lifecycle carbon performance of digital infrastructure (Wang et al. 2025). More granular and output-linked metrics are essential for informed decarbonization strategies and investment decisions (Springer Nature 2025).

Global and Regional Context

Efforts to decarbonize data centers are evolving globally, influenced by distinct regional regulatory approaches, technological advancements, and market dynamics. Key regions like the United States, Europe, and China provide valuable context for understanding the challenges and strategies relevant to the GCC.

Quantifying the escalating global energy demand underscores the urgency of decarbonization. In this context, Oxford Energy Studies (2025) developed a statistical model based on historical data center electricity consumption, correlating it with variables like GDP, population, trade, and climate. Calibrated against historical efficiency improvements and the rise of hyperscale facilities see Figure 1 and 2, their projections offer a relatively conservative estimate compared to some industry forecasts. Figure 2 also shows an exponential electricity consumption forecast in certain regions and countries.

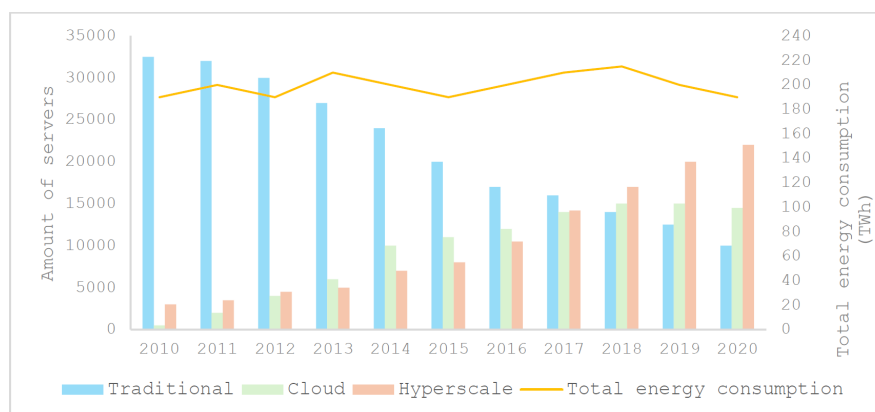


Figure 1—Historical Energy Consumption of data centers globally (Oxford Studies 2025)

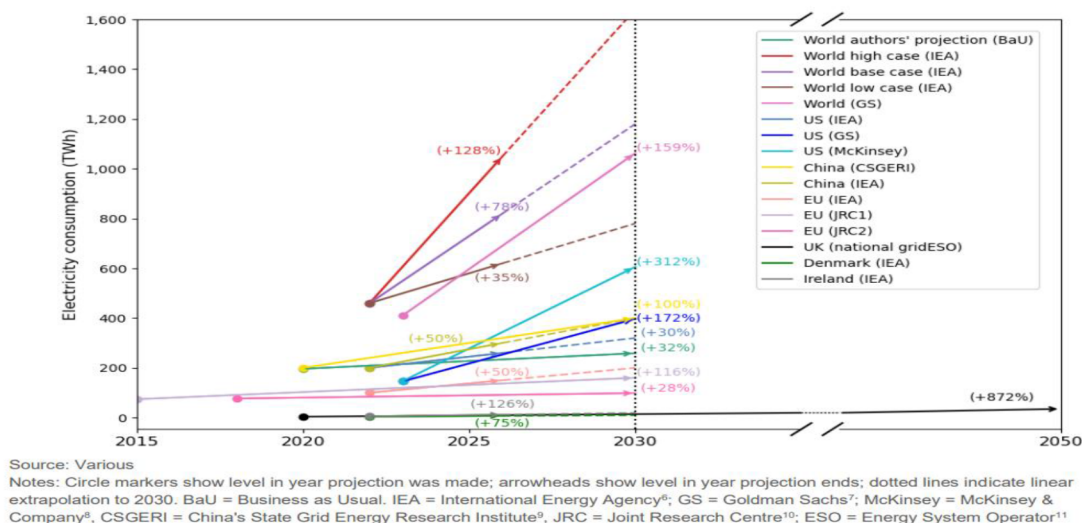


Figure 2—Projected increase in data centers electricity consumption in different regions (Oxford Studies 2025)

United States

Figure 3 shows how the US is the leader in data center innovation, with significant investments in energy efficiency and renewable energy. Regulations such as the Energy Star program and state-level incentives have driven the adoption of green practices.

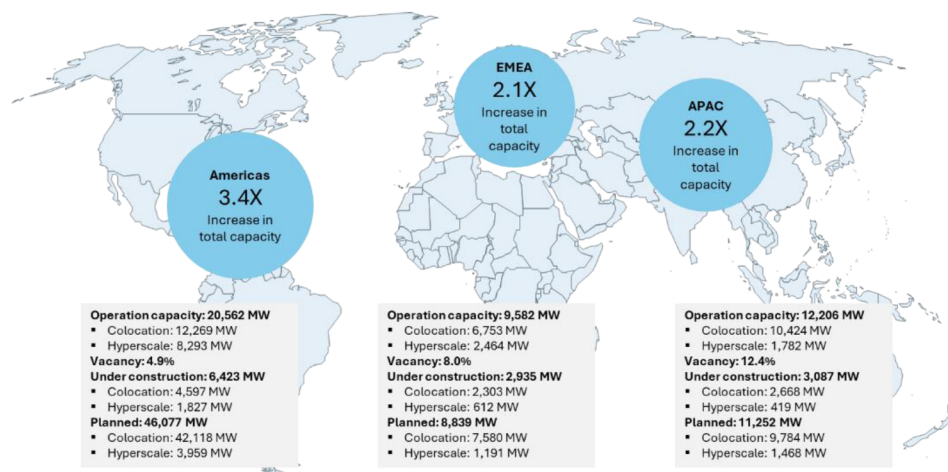


Figure 3—Regional Growth (Cushman & Wakefield 2025)

At the same time, the US plays a central role in shaping the global trade of AI chips, hardware that has become critical to modern data centers. This is driven by the rise of generative AI and large language models (LLMs) have dramatically increased the need for Graphic Processing Units (GPU), which excel at parallel processing tasks for training these models. As AI applications expand beyond IT into smart devices and edge computing, the demand for advanced semiconductors will only accelerate (Baker 2025). As an example, Taiwan's exports of semiconductors—including integrated circuits and data processing units - to the United States surged between February and June 2025, reaching a record \$11.5 billion (Soltani 2025).

To manage this strategic technology, the Biden administration introduced export controls targeting advanced AI chips. Under the Framework for Artificial Intelligence Diffusion, countries are classified into three tiers. Tier 1 includes the US and 18 key allies with unrestricted access. Tier 2 covers most other nations, including India, Israel, the UAE, and Saudi Arabia, which face controlled access with licensing requirements and caps on AI chip imports (Baker 2025). However, the Department of Commerce's Bureau of

Industry and Security (BIS) formally rescinded the Framework for Artificial Intelligence Diffusion interim final rule adding further scrutiny to data centers that house advanced computing integrated circuits used to train AI models for or on behalf of entities headquartered in countries of concern (Crowell 2025). More recently, Bloomberg anticipated that the Trump administration is working on a new regulatory update in order to reduce what it considers unnecessary regulatory barriers, promoting innovation and adoption, and supporting a global deployment of American AI technologies, which is still unclear at the time of this paper's publication (Seddiq et al. 2025). It should be noted that as of August 2025, the rescindment rule has not been published in the US Federal Register, and therefore, the AI Diffusion Rule is still legally in effect.

Amid these regulatory shifts, the US data center sector is experiencing explosive growth in power demand, driven heavily by AI. In fact, power demand from data centres in the United States is projected to make up 9 % of total electricity demand by 2030, according to the Electric Power Research Institute (2025). Oxford Energy Studies (2025) based on other studies estimates this figure will reach almost 12% by the same year, representing a 17.9 percent compound annual growth rate relative to 2023. Figure 4 below shows the data-centre-driven demand increases (Oxford Energy Studies 2025)

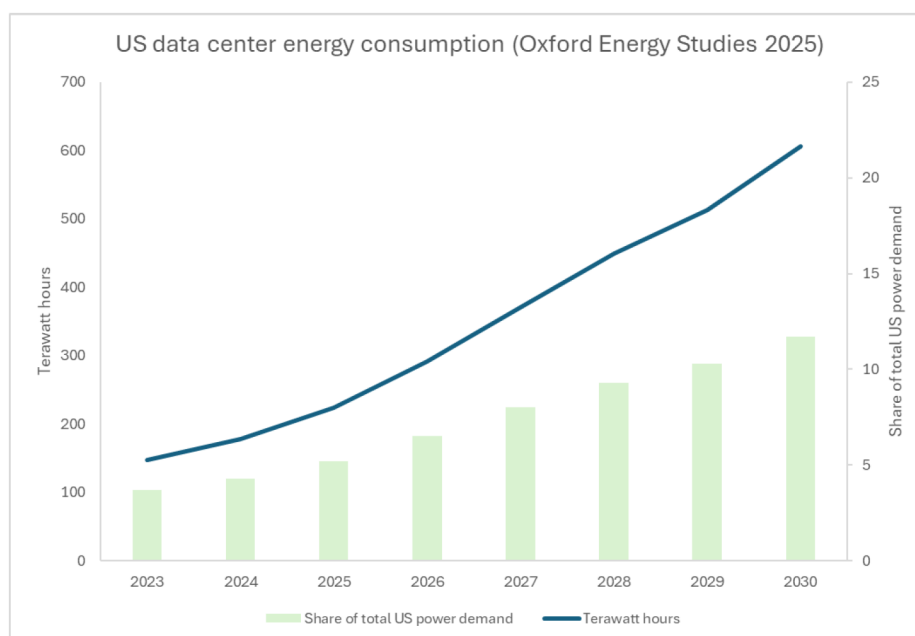


Figure 4—Projected increase in DC electricity consumption in different regions (Oxford Energy Studies 2025)

Europe

Europe has taken a regulatory-driven approach to promote data center energy efficiency and sustainability, with both EU-wide and national initiatives setting high standards. Key initiatives include the EU Code of Conduct for Data Centre Energy Efficiency, encouraging best practices, and the mandatory requirements of Germany's Energy Efficiency Act. At the EU level, the Climate Neutral Data Centre Pact, aligned with the European Green Deal (European Commission 2025ab), commits hyperscalers and colocation providers to ambitious sustainability targets, such as achieving a PUE of 1.3 to 1.4 by 2025, sourcing 75–100% of electricity from renewables by 2025 and 2030, respectively, and optimizing water usage.

The UK government has launched the AI Energy Council (2025), a cross-sector initiative aligning AI-driven infrastructure growth with clean energy goals. The council includes major technology firms and energy providers, aiming to accelerate grid connections to renewable and nuclear sources, coordinate infrastructure upgrades, and develop AI Growth Zones. However, delays in grid connection have raised concerns about increased reliance on gas-fired backup capacity, potentially undermining decarbonization goals (UK Government 2025). These initiatives reflect Europe's strategic effort to ensure that the growth

of data center and AI infrastructure is supported by strong sustainability standards and alignment with net-zero ambitions.

China

China's rapid digital economy expansion has spurred massive investment in data center infrastructure. Recognizing the associated energy impact, the government launched the "Eastern Data, Western Computing" initiative. This national strategy aims to relocate data processing capacity from energy-constrained eastern regions to western areas rich in renewable energy resources, seeking to optimize national energy use and promote greener data center operations across the country ([State Council Information Office 2024a](#)). By March 2024, over 1.46 million standard server racks had been installed across 10 national data center clusters, reflecting the scale and ambition of the project ([State Council Information Office 2024b](#)).

Regional context: GCC

The GCC countries have started to develop regulatory frameworks to promote the decarbonization of data centers. These regulations focus on energy efficiency, renewable energy integration, and environmental sustainability. The [figure 5](#) shows the number of data centers in pre-execution and execution phase in the region.

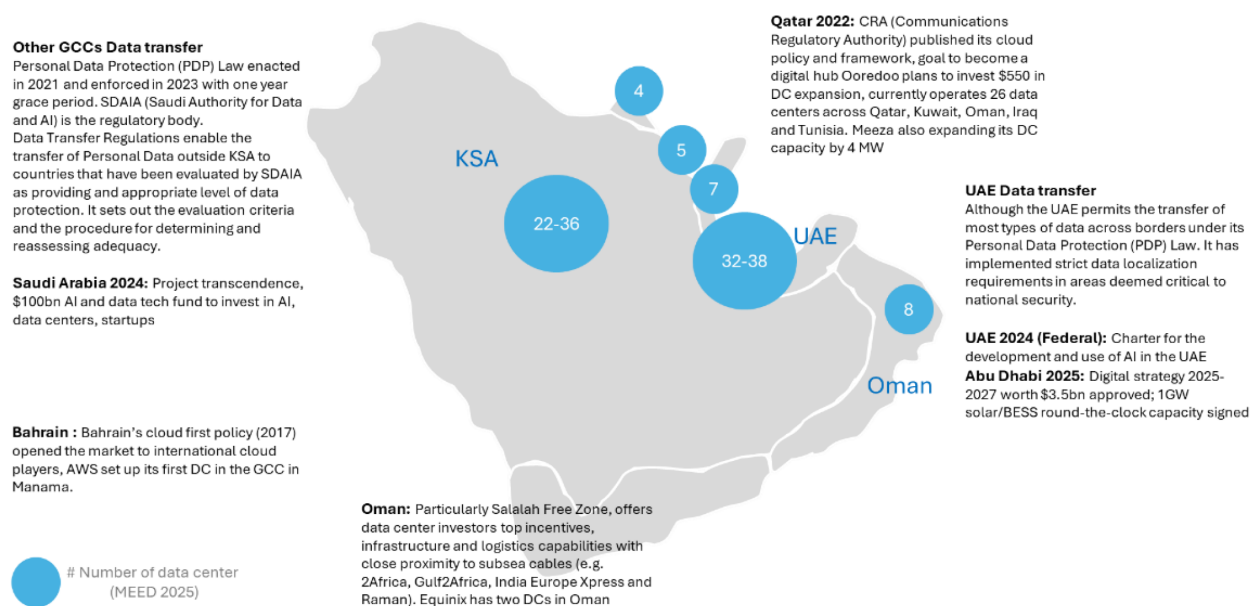


Figure 5—GCC countries – number of Data Centers and regulations (MEED 2025)

The UAE has introduced groundbreaking -Federal Decree-Law No. (11) of 2024, aimed at reducing the effects of climate change, which is applicable to all the UAE companies, including Data Centers. Effective May 30, 2025, this law sets out requirements for emissions reduction, ecosystem protection, and the promotion of sustainable innovation, positioning the UAE as a leader in global climate action. Fines will be imposed for non-compliance, with repeated violations leading to higher penalties can achieve the amount of AED for each category of non-compliance. This emphasizes the need for strict adherence to the law for critical infrastructure and security. ([BDO 2024](#)). The UAE's cybersecurity framework, managed by the Telecommunications and Digital Government Regulatory Authority (TDRA) and Dubai Electronic Security Center (DESC), includes the Cybersecurity Law (Federal Decree-Law No. 34/2021) and emirate-specific policies like the Dubai Cybersecurity Strategy which reflects in complex licensing requirements.

The Kingdom of Saudi Arabia (KSA) is poised to capitalize on the burgeoning demand for data centers. This surge in demand represents a significant opportunity as the market remains underserved — KSA's data

center capacity is 50-80% lower than leading regions such as the EU, US, China, Singapore, and the UAE. KSA's proactive approach to creating a conducive regulatory environment is about attracting investment and fostering innovation. By implementing frameworks like the Cloud Computing Regulatory Framework (CCRF) and the Personal Data Protection Law (PDPL), the Kingdom is signalling its commitment to data privacy and laying down a solid foundation for sustainable digital growth. This also indicates a forward-thinking approach that prioritizes economic development and ethical considerations. (Berger 2025)

The KSA Data Center regulations are issued by the Communications, Space, and Technology Commission (CST), these regulate registration, energy management, and sustainability to foster a competitive sector. KSA also has cybersecurity requirements which are set out by the National Cybersecurity Authority (NCA), ECC-2 mandates cybersecurity governance, data localization, and Saudization. The KSA Vision 2030 and Green Saudi Initiative require energy efficiency and renewable energy adoption.

In the case of Bahrain, The Personal Data Protection Law No. 30 of 2018 (PDPL) was issued in 2018 in the Kingdom of Bahrain. The legislation was directly influenced by the country's ambitious plans to become a hub for data centers, with Amazon Web Services (AWS) planning on opening massive data centers there by 2019 (End of Point Protector 2025).

GCC Capabilities Mapping

The following map in figure 6 showcases the grid vulnerability and potentialities in the GCC countries, including renewables, nuclear, CO₂ storage and fibre optic network.

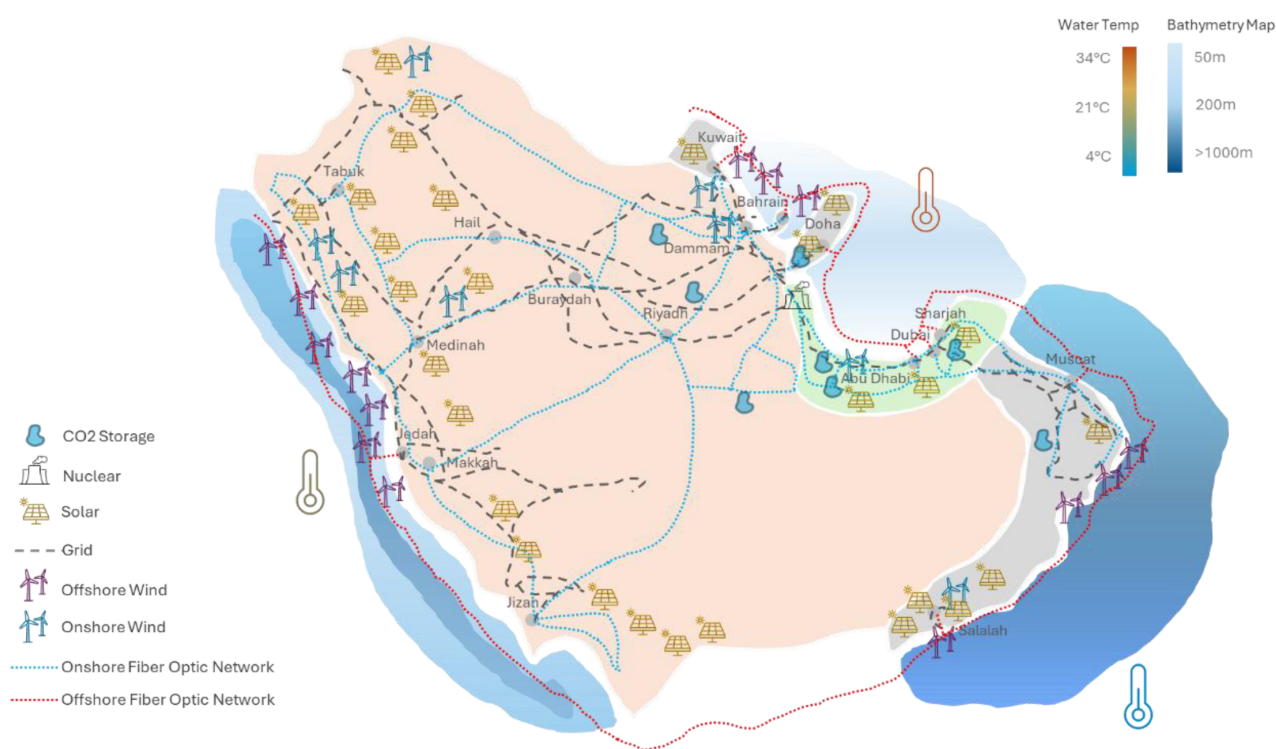


Figure 6—GCC Capability Mapping (Sherif 2025; MEES 2020; GENI 2025; Mahdy et al. 2025a; Mahdy, et al. 2025b; Bahaj et al. 2025; Mostafa et al. 2025; World Bank 2025; TechXmedia 2025; Solar GIS & World Bank 2025; L'Hegaret et al. 2025; Raitos et al. 2025; Rezaei-Latifi 2025; Thatti 2025)

The composite view is based on five dimensions: (i) grid interconnection and redundancy, (ii) firm low-carbon capacity (e.g., nuclear generation), (iii) renewable energy potential (solar and wind), (iv) proximity to CO₂ storage basins, and (v) fiber-optic backbone density. Each dimension was normalized on a 0–100 scale using min–max normalization across the GCC, then combined using indicative weights: grid/interties (30%), firm low-carbon supply (25%), renewables potential (20%), CO₂ storage proximity (15%), and fiber

connectivity (10%). The resulting composite score is represented by the color gradient, while icons overlay individual resource layers for quick reference. Under this framework, the UAE ranks highest due to its strong grid interconnections, operational nuclear capacity, large-scale solar build-out, and dense fiber network, as documented in the cited sources.

Technical Solutions for Decarbonization

Data Centers are significant consumers of energy, primarily due to their requirement for continuous operation and the energy-intensive cooling systems needed to maintain optimal equipment temperatures. This substantial energy demand contributes significantly to GHG emissions, underscoring the need for effective technical solutions to reduce environmental impact ([Smith 2023b](#)). Key strategies include the integration of carbon capture and storage (CCS), renewable energy sources, cooling optimization, and direct air capture (DAC) to mitigate emissions associated with both on-site operations and upstream electricity generation ([Jones 2022a](#)).

Improving the energy efficiency of cooling infrastructure, through advanced thermal management, liquid cooling, or AI-driven controls, can significantly lower energy usage and emissions ([Brown 2021a](#)). Additionally, innovative approaches such as waste heat recovery can further support sustainability by repurposing excess thermal energy for use in other applications in integrated systems, such as in district heating or nearby industrial processes ([Green 2024a](#)). Together, these combined efforts are essential for creating more sustainable and environmentally friendly data centers ([White 2025a](#)).

Technology Options For Decarbonizing Emissions and Power Consumption for Data Centers

CCS & DAC

While traditionally associated with heavy industry and power generation, Carbon Capture and Storage (CCS) emerge as a potential solution for reducing emissions from data centers, particularly those relying on on-site backup generators or grid electricity derived from fossil fuels. CCS can be applied to capture CO₂ emissions from combustion sources within the facility, such as diesel generators, enabling localized mitigation of operational emissions ([Smith 2023c](#)). Integration with modular, containerized capture units allows retrofitting without major infrastructure redesigns ([Jones 2022b](#)).

Direct Air Capture (DAC), by contrast, removes CO₂ directly from ambient air, independent of emission source. This makes it a flexible solution for data centers, especially those aiming for net-zero or carbon-negative operations ([Brown 2021b](#)). When powered by renewable energy or coupled with waste heat from IT or cooling systems, DAC systems can achieve higher operational efficiency and offer a meaningful pathway for offsetting residual emissions ([Green 2024b](#)).

Although both CCS and DAC technologies currently face economic and scalability challenges, ongoing innovation, regulatory incentives, and integration with carbon markets are driving increased feasibility ([White 2025b](#)). As data centers seek comprehensive decarbonization strategies, the co-location or partnership with carbon capture or cluster infrastructure may become a viable component of long-term sustainability planning ([Black 2023](#)).

Renewables, Batteries and Storage

As data centers expand rapidly in the Middle East and other high-temperature regions, efficient and reliable cooling systems are becoming critical drivers of operational cost and uptime. The interplay between cooling systems and electrical infrastructure directly influences a data center's reliability and total cost of ownership. In these challenging climates, energy-efficient cooling solutions, such as indirect evaporative cooling, liquid cooling, or thermal storage, are increasingly prioritized to ensure thermal resilience while minimizing energy use ([Smith 2023d](#)). Strategies for cooling optimization, such as the combined use of free-cooling

and economizers configuration by using deep sea cooling water systems, help to reduce the energy demand of the data center cooling system (Jones 2022).

At the same time, renewable energy adoption is reshaping the data center industry. The integration of solar, wind, hydroelectric, and geothermal power enables data centers to significantly reduce their carbon footprint while supporting broader energy transition goals (Brown 2021a). This shift not only aligns with corporate sustainability commitments but also enhances brand reputation and stakeholder confidence (Green 2024). Additionally, innovations such as heat reuse, where excess heat from IT equipment is captured and repurposed, further contribute to energy efficiency and sustainability (White 2025c). As data centers continue to integrate renewable energy and advanced cooling technologies, they are setting new standards for environmental stewardship and operational excellence (Black 2023).

In this context, renewable energy integration is becoming a central pillar of data center design. Photovoltaic (PV) systems and wind turbines, deployed either on-site or via power purchase agreements (PPAs), provide clean, reliable electricity and reduce dependency on fossil-fuel-based grids (Smith 2023d). The Middle East, with its high solar irradiance and expanding renewable infrastructure, offers particularly strong potential for solar PV deployment (Jones 2022c). Wind energy, while more site-specific, complements solar by providing generation diversity (Brown 2021a).

Hydroelectric power, where available, adds further flexibility. Reservoir-based hydro systems can offer dispatchable power to help stabilize supply during intermittent solar or wind output, while run-of-river hydro provides steady baseload generation (Green 2024). In suitable regions, geothermal energy contributes clean, round-the-clock power. Technological advancements—such as enhanced geothermal systems and closed-loop drilling—are expanding access to geothermal energy even in locations beyond traditional volcanic zones (White 2025c). These systems can also provide efficient direct-use heat exchange for cooling applications (Black 2023).

To balance renewable intermittency and support grid stability, Battery Energy Storage Systems (BESS) are increasingly used to store excess energy for use during peak demand or outages. These systems also enable hybrid microgrids that improve energy resilience, support load shifting, and participate in demand response programs (Smith 2023d).

Looking ahead, from the perspective of energy security as an important pillar of the energy trilemma, emerging technologies like small modular reactors (SMRs) are gaining attention as reliable, zero-carbon power sources. Particularly in the Middle East, where nuclear energy is part of national diversification strategies, SMRs could play a role in powering future data centers with consistent, high-density electricity while supporting climate goals (Jones 2022c).

Energy Efficiency Measures & Sustainability

Energy efficiency in data centers can be enhanced through several key strategies. Advanced cooling technologies such as liquid cooling, free cooling, and economizers significantly reduce energy consumption, with liquid cooling being more efficient than traditional air methods (Pears and Shields 2019). Upgrading to energy-efficient IT equipment and employing virtualization and server consolidation also help minimize power usage. Monitoring and improving PUE is essential for identifying energy-saving opportunities. Sustainable design practices, including the use of green building materials like recycled steel and carbon-capture concrete, along with modular construction, reduce environmental impact during development (Pears and Shields 2019). Circular economy practices such as robust e-waste management and conducting lifecycle assessments (LCA) further support sustainability by promoting resource recovery and informed decision-making.

Offshore Solution, Repurposing Existing Offshore Facilities & Subsea

Vessel Repurposing

Repurposing vessels such as ships or oil rigs into floating data centers offers several advantages, including mobility, scalability, and proximity to renewable energy sources such as offshore wind and solar power. These floating platforms can utilize seawater for cooling, significantly reducing energy consumption and enhancing thermal efficiency. This method leverages the natural thermal properties of water to maintain optimal temperatures for IT equipment. Additionally, floating data centers reduce the need for land-based infrastructure, thereby minimizing environmental impact and land use ([Martin and Peskoe 2025](#)).

Subsea Data Centers

Underwater data centers represent a promising innovation in the pursuit of energy efficiency and sustainability. Microsoft pioneered this concept with Project Natick, deploying its first large-scale underwater data center in 2015. The project demonstrated that subsea data centers could improve energy efficiency, reduce cooling costs, and enhance reliability by minimizing human interference and exposure to environmental fluctuations ([Microsoft 2020](#)).

The International Energy Agency (IEA) has warned that global electricity consumption by data centers could more than double from 460 TWh in 2022 to over 1,000 TWh by 2026, making energy-efficient cooling solutions a top priority. Since cooling accounts for 25-40% of total energy use in data centers, innovations such as two-phase immersion cooling and seawater-based systems are critical ([Martin and Peskoe 2025](#)).

Subsea data centers benefit from a virtually limitless supply of cold water for cooling, resulting in excellent Water Usage Effectiveness (WUE). They also offer access to renewable energy sources like tidal, wave, and offshore wind, supporting the development of green data centers. Their proximity to coastal population hubs reduces latency and improves data transmission speeds ([Microsoft 2020](#)). Moreover, these facilities demonstrate improved reliability and lower failure rates. For example, Microsoft's 2018 experiment showed that server failure rates in its underwater data center were significantly lower than those in traditional land-based facilities. Modular designs, reduced maintenance needs, and enhanced physical security due to restricted access further contribute to their appeal ([Microsoft 2020](#)).

Technological advancements in subsea data centers include pressure-resistant enclosures, autonomous operation systems, and advanced fibre optic connectivity. Real-time environmental monitoring and the use of durable materials ensure operational efficiency and longevity. However, challenges remain, including accessibility for maintenance, environmental impact assessments, and compliance with marine regulations ([Subsea Cloud 2025](#)). Despite these challenges, subsea data centers offer compelling economic and environmental benefits. They reduce cooling-related energy costs, minimize hardware failures, and support modular, scalable deployment. As more companies explore this model, the potential for innovation and sustainable growth in the data center industry continues to expand ([Microsoft 2020](#); [Martin and Peskoe 2025](#); [Subsea Cloud 2025](#)).

Options for Using Energy from Data Centers

Waste Heat Recovery

Cooling in data centers is fundamentally about removing waste heat. As the industry approaches the limits of Power Usage Effectiveness (PUE) improvements, attention is shifting toward managing this waste energy more effectively. Cooling methods vary by climate: free cooling is common in colder regions using ambient air or water sources, while warmer climates rely on chillers with cooling towers or condensers. Liquid immersion cooling, where servers are submerged in dielectric fluid, offers significantly higher efficiency and allows for greater server density. Despite these advancements, most waste heat is still expelled from the facility and remains unused ([The Green Grid, 2021](#)).

To address this inefficiency, innovative heat recovery strategies are emerging. District heating systems can repurpose excess heat from data centers to warm nearby buildings, offering both environmental and economic benefits. Another approach is heat-to-power conversion, where waste heat is transformed into electricity. These systems often rely on heat pumps, which transfer heat from the data center's exhaust to district heating networks. This integration enhances the performance of both systems, reduces CO₂ emissions, and supports broader sustainability goals (Apostolopoulou 2025).

Efforts to reduce data center energy consumption focus on improving IT equipment and optimizing cooling systems. Cooling improvements include alternative sources like air-side free cooling and energy efficiency measures such as waste heat recovery (WHR). However, WHR faces challenges due to the low-grade nature of the heat, typically between 20°C and 55°C, which requires additional energy input via heat pumps. Successful implementation depends on integrated planning and reliable heat utilization networks, making WHR a promising but complex area for innovation in sustainable data center operations (Apostolopoulou 2025), see Figure 7.

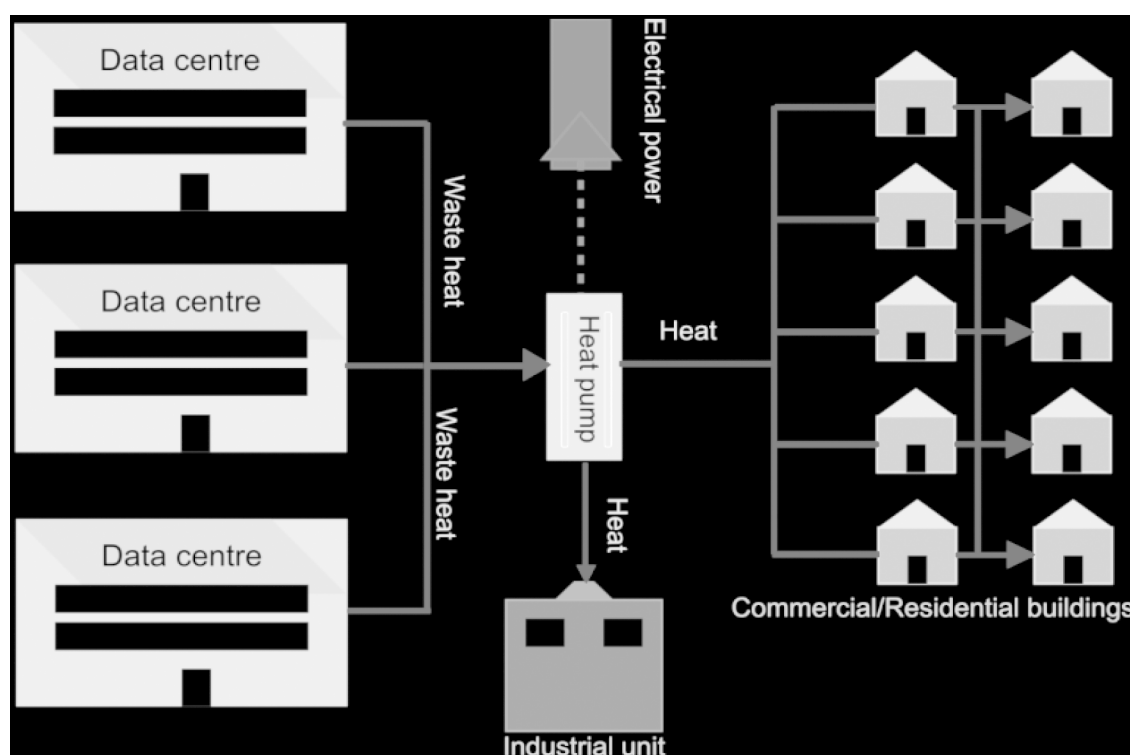


Figure 7—WHR for Use in Heat Networks (Apostolopoulou 2025)

Technology Comparison

Data centers present a dilemma in the context of decarbonization. They are essential for technological advancements that reduce carbon footprints, such as optimizing operations and supporting renewable energy innovations. However, their high energy consumption, necessary to power servers and cooling systems, can lead to significant carbon emissions. This paradox means that while data centers can aid decarbonization, their energy intensity can also counteract it. Addressing this requires improving energy efficiency, integrating renewable energy sources, and adopting sustainable practices within data center operations, energy efficiency measures, and carbon capture technologies. These solutions can be classified based on their impact and feasibility.

To provide a holistic view of decarbonization options for data centers (DC) in the GCC region, the following tables merge the technical, cost, efficiency, and operational details into the regional context of Table 2 (Onshore) and Table 3 (Offshore).

Table 2—Consolidated Onshore Power Supply Solutions for GCC DC Projects

Solution	Benefit	Drivers / Advantages	Disadvantages	Costs	Efficiency	Independency	Flexibility	Reliability	Potential GCC Country
Renewables + Battery (Hussain, 2025 and Goldman Sachs 2025)	Proven tech, reduced carbon footprint, reliable clean energy	Regulations, regional ambitions, energy independence	Land requirement, intermittent power, capacity limits, intermittency issues	Moderate, high initial CAPEX, low OPEX	Energy efficiency with storage	High	Moderate	Subject to intermittency	KSA, UAE, Oman
CCU + Waste energy / DAC / Carbon Capture & Storage (CCS) (Han, et al. 2025 and Goldman Sachs 2025)	Emission reduction, utility application, proven technology	Net zero ambitions, heat recovery options if integrated with DC, flexibility	High up-front cost, no regional carbon mechanisms, long-term storage risks, regulatory complexity	High CAPEX, energy intensive OPEX	Heat recovery can improve efficiency	Depends on infrastructure	Applicability to existing systems	Reliant on storage and verification	KSA, UAE, Qatar
Nuclear (SMR) (Wilson 2024)	Reliable, minimal carbon footprint	Small land footprint, energy independence	Supply chain, waste, labor, uranium supply, long lead time, public acceptance	High CAPEX & OPEX	High	High	Limited flexibility	High reliability	UAE
Gas / LNG (Hussain, 2025)	Reliable, abundant resources	Greener than oil/ coal, small footprint	Lower carbon efficiency, interconnection delays	Moderate	Moderate	High	Moderate	High	Qatar, KSA, UAE, Oman, Bahrain
Energy Efficiency / Advanced Cooling Technologies (Han, et al. 2025 and Goldman Sachs 2025)	Quick wins, energy savings	Lower costs, proven options, improved performance	Limited total savings, compatibility and maintenance requirements	Low to moderate, high initial investment	High	Moderate	Improved performance	High	All
Virtualization & Server Optimization (Wilson 2024)	Cost reduction, greater efficiency	Scalability, applicable to existing systems	Compatibility issues with legacy infrastructure	Initial investment, long-term OPEX savings	Increased efficiency	Neutral	High	Depends on IT resilience	All
Carbon Offsetting (Hussain, 2025)	Immediate impact for emission targets	No initial investment required	Temporary solution, verification challenges	Ongoing costs	N/A	None	Flexible	Relies on third-party schemes	All

Table 3—Consolidated Offshore Power Supply Solutions for GCC DC Projects

Solution	Benefit	Drivers / Advantages	Disadvantages	Costs	Efficiency	Independency	Flexibility	Reliability	Potential GCC Country
Renewables + Battery (Coyle 2025 and Zhang 2019)	Proven tech, reduced carbon footprint, reliable clean energy	Regulations, regional ambitions, energy independence	Intermittent power, capacity limits, intermittency issues	Moderate CAPEX, low OPEX	High with sufficient battery	High	Moderate	Intermittency risk	KSA, Oman
Offshore / Subsea Cooling Systems (Coyle 2025 and Zhang 2019)	Significant energy savings, utilizes water resources	Access to abundant water for cooling, improved efficiency	Logistics, environmental impact, sea temperature variation, high CAPEX	High initial investment, moderate OPEX	High	Neutral	Improved performance	Site-dependent reliability	Oman, KSA
Energy Efficiency (Kamiya et al, 2035)	Quick wins, energy savings	Lower costs, proven options	Limited carbon savings	Low	High	Moderate	Improved performance	High	All
Advanced Cooling Technologies (Kamiya et al, 2035)	Enhanced energy savings and reliability offshore	Compatibility with offshore environments	Maintenance and specialized infrastructure needed	High	High	Neutral	Improved performance	Depends on system design	All
Virtualization & Server Optimization (Kamiya et al, 2035)	Cost reduction, greater efficiency	Scalability, applicable to existing systems	Compatibility issues with legacy infrastructure	Initial investment, long-term OPEX savings	High	Neutral	High	Depends on IT resilience	All
Carbon Offsetting	Immediate impact for emission targets	No initial investment required	Temporary solution, verification challenges	Ongoing costs	N/A	None	Flexible	Relies on third-party schemes	All

Note: Items such as Nuclear and LNG are generally not applicable to offshore solutions but are included in onshore scenarios. The consolidated tables ensure a comprehensive view by integrating technology options, operational implications, and regional relevance for decision-makers in the GCC region.

Decarbonization Index for Data Centers

A decarbonization index was developed to measure the progress of data centers in reducing their carbon footprint. This index would consider factors such as energy efficiency, renewable energy adoption, and carbon capture technologies.

Sustainable Digital Efficiency Index (SDEI)

As data centers continue to grow in strategic and economic importance, it is increasingly necessary to evaluate both their financial and environmental performance in tandem. Financial viability - particularly capital efficiency and cost of capital - is essential for securing funding and supporting long-term growth. At the same time, environmental performance, especially energy use and carbon emissions, is critical for ensuring regulatory compliance, cost competitiveness, and alignment with stakeholder sustainability expectations.

To address this need, the Sustainable Digital Efficiency Index (SDEI) provides a single, integrated measure for evaluating the carbon efficiency, decarbonisation readiness, and bankability of digital infrastructure projects. By combining these three components into a composite score, the SDEI offers investors, operators, and policymakers a transparent and consistent framework for comparing data center projects and guiding investment decisions. Evaluating these dimensions together supports informed decision-making, reduces investment risk, and promotes resilience in a decarbonising global economy.

$$SDEI = CES_{norm} \times W_1 + (DRS_{norm} \times W_2) + (B_{norm} \times W_3)$$

Where:

CES_{norm} (Carbon Efficiency Score): A normalized score where *lower carbon intensity = higher score*. It reflects how efficiently a data center converts energy into digital output (e.g., tokens), capturing lifecycle carbon intensity.

DRS_{norm} (Decarbonization Readiness Score): A normalized score where *higher = better*. It reflects the maturity of decarbonization strategies such as renewable energy use, carbon offsets, and infrastructure readiness.

B_{norm} (Bankability Score): A normalized score where *higher represents more bankable*. It reflects financial viability, including lifecycle cost efficiency, carbon performance, and access to green or preferential financing.

Recommended Weights: Weightings assigned to each component (default: $W_1=0.3$, $W_2=0.3$, $W_3=0.4$).

Interpretation: A composite score ranging from 0-100 that enables consistent and meaningful comparisons across data center projects, development scenarios, and policy environments.

The SDEI framework builds on composite scoring methodologies commonly used in infrastructure and energy project assessments. Its additive, weighted structure mirrors traditional bankability models, where financial, technical, and ESG dimensions are integrated to inform investment decisions.

Higher SDEI scores indicate stronger sustainability performance, reflecting lower lifecycle carbon intensity (CES), greater decarbonization readiness (DRS), and stronger financial attractiveness (B). By normalizing and weighting these parameters, the SDEI enables flexible yet rigorous comparisons across data centers, development scenarios, and policy environments – supporting more sustainable and bankable digital infrastructure development.

SDEI Component Scores: CES, DRS, and Bankability

The Carbon Efficiency Score (CES) measures the energy and carbon efficiency of data center operations relative to the amount of digital output they produce. It reflects how effectively a facility converts energy into digital services, while accounting for both operational and embodied carbon.

For Large Language Models (LLMs) and AI models, tokens represent the fundamental unit of computation and user interaction. Using tokens as the output metric provides a granular and meaningful basis for evaluating carbon efficiency across different workloads and facilities.

Formula (conceptual):

$$CES = \frac{TotalTokensProcessed}{TotalEnergyConsumption(kWh) \times TotalEmissionsFactor(kgCO_2e/kWh)}$$

Total Energy Consumed: All energy used by the data center (or AI cluster).

Emissions Factor: Includes both operational emissions (grid mix) and allocated embodied emissions per kWh over the system's lifetime.

Tokens Processed: Total number of tokens processed for inference and/or training.

Interpretation: Higher CES values indicate better carbon efficiency, meaning the facility delivers more digital output per unit of carbon emitted. By focusing on carbon intensity per token delivered, CES enables meaningful comparisons across data centers or AI clusters of varying sizes and workloads.

Decarbonization Readiness Score (DRS)

The Decarbonization Readiness Score is a forward-looking measure of how actively a facility is implementing decarbonization levers such as renewable energy adoption, efficiency improvements, and emissions offset strategies. DRS complements CES by indicating a facility's future potential to improve sustainability performance.

Formula (weighted score approach):

$$DRS = (W_1 \times \text{RenewableEnergyUse}) + (W_2 \times \% \text{EnergySavings from Efficiency Projects}) \\ + (W_3 \times \% \text{GHG Offset or Capture}) + (W_4 \times \% \text{CUE Improvement})$$

Renewable Energy Use (%): Share of electricity sourced from renewable energy.

Energy Savings from Efficiency Projects (%): Annual energy consumption reduction relative to a baseline year.

GHG Offset or Capture (%): Proportion of annual operational emissions mitigated via verified offsets or on-site carbon capture.

CUE Improvement (%): Reduction in Carbon Usage Effectiveness (CUE) compared to baseline year.

Recommended Weights: $W_1 = 40\%$, $W_2 = 30\%$, $W_3 = 20\%$, $W_4 = 10\%$.

Interpretation: 0 to 100 – higher scores indicate stronger decarbonization performance and readiness.

Classification Example:

80–100: Advanced – Fully integrated decarbonization strategy.

60–79: Progressive – Strong renewable and efficiency mix.

40–59: Transitional – Partial implementation, limited renewables.

<40: Lagging – Minimal sustainability interventions.

Bankability Score (B)

As data centers become the backbone of the global digital economy, their financial and environmental footprints are drawing increased scrutiny. From an investment perspective, bankability reflects a project's ability to secure financing by demonstrating predictable cash flows, acceptable risk-adjusted returns, and alignment with government regulatory and policy frameworks including ESG, for all stakeholders. For data centers, decarbonization strategies, such as renewable energy integration, advanced thermal management, and demand-side optimization, are increasingly critical for maintaining competitiveness and mitigating exposure to carbon pricing, regulatory constraints, and energy market volatility.

Through this analysis, we aim to establish how decarbonized data centers can become a bankable, sustainable infrastructure asset, capable of attracting both conventional and green financing in global and GCC markets.

The benefits of decarbonizing data centers can be analysed from two complementary perspectives: (a) the external, system-level impacts on energy systems and environmental sustainability, and (b) the internal, operator-level advantages that directly enhance data center performance (in terms of energy and emissions reduction) and financial outcomes. At the system level, decarbonization strategies such as renewable generation, energy storage, and demand-side optimization can reduce stress on local grids and, when coordinated effectively, strengthen system resilience and stability (IEEE Spectrum 2019). Simultaneously, within the data center, measures such as advanced thermal management, demand-side optimization, and energy-efficient IT infrastructure drive operational efficiency. Renewable integration, while essential for reducing emissions, primarily affects the carbon intensity of power consumed rather than internal energy efficiency (IEEE Spectrum 2019).

The data center industry delivers substantial economic benefits through revenue generation and investment, supporting critical digital infrastructure that enables innovation and business continuity (Statista 2025). However, the rising energy demand presents challenges, including higher operational costs and environmental concerns. Addressing these issues requires continued focus on energy efficiency improvements and the adoption of renewable energy sources to ensure sustainable growth (Statista 2025).

To quantitatively assess the bankability of decarbonized data centers (Nagarathinam et al. 2024), it is useful to establish a framework that links financial performance with operational efficiency and environmental impact. The following equation introduces a Bankability Score (B) that integrates key variables influencing investment attractiveness:

$$\text{Bankability Score}(B) = \left[1 - \frac{\text{CapEx} + \sum_{t=1}^T \frac{\text{OpEx}_t}{(1+r)^t}}{\sum_{t=1}^T \frac{R_t}{(1+r)^t}} \right] \times \frac{WACC_{\text{baseline}}}{WACC_{\text{project}}}$$

R_t : Projected revenue stream in year t .

CapEx : Lump-sum capital expenditure incurred at time $t = 0$. This model assumes a one-time cost covering all upfront investment requirements (e.g., land acquisition, site preparation, facility construction, initial equipment purchases), and it is not distributed over multiple years.

OpEx_t : Operating expenditures incurred annually over the asset's lifecycle.

t : Time index representing lifecycle years of operation.

r : Discount rate used to calculate present value.

$WACC_{\text{baseline}}/WACC_{\text{project}}$: Weighted Average Cost of Capital, a financing ratio, which captures external influences on bankability such as access to preferential financing, regulatory incentives, or green financing. It reflects how favorable financing terms improve project viability.

This formulation mirrors the way infrastructure investors assess real-world projects: using discounted cash flows to account for time value of money, repayment capacity, and long-term financial viability. Rather than treating CapEx and OpEx as static or evenly distributed costs, this approach evaluates their impact over time—making it better suited for comparing investment readiness across diverse project types and financing environments.

Traditional cost metrics such as Total Installed Cost (TIC) focus solely on upfront capital requirements which is useful for development-stage budgeting as it captures the upfront capital required for land, construction, and infrastructure, but insufficient for understanding a project's overall financial viability. Similarly, Total Ownership Cost (TOC) includes lifecycle operating expenditures but may obscure near-term cash flow considerations by distributing costs evenly across long project horizons (Nagarathinam et al., 2024). In contrast, the Bankability Score addresses this gap by using NPV of cash flows and revenues, providing a more realistic and investment-aligned view. It enables early screening and prioritization of projects, especially when quick decisions must be made across portfolios with varying technical scope, cost structures, and risk exposure.

Preferential financing mechanisms—such as green bonds, concessional loans, tax credits, and sovereign wealth fund investments—can significantly lower the cost of capital for data center projects. This results in a financing ratio greater than one, indicating improved financial attractiveness due to reduced debt servicing costs and risk premiums. In contrast, projects without such access may face higher financing costs, with a financing ratio below one. Incorporating the financing ratio into project evaluation highlights how decarbonization strategies not only enhance operational performance but also unlock more favourable financing, which can be a decisive factor in investment decisions (Baker Botts 2025).

These financing advantages can amplify the appeal of decarbonized data centers by improving return profiles and reducing capital costs. For example, district heating integration or renewable energy use may align with national sustainability goals, making projects more eligible for green financing. However, the absence of such incentives can diminish investor confidence, even if operational efficiencies are present. This underscores the importance of aligning data center projects with broader policy frameworks to maximize their bankability.

Despite its usefulness, the financing ratio has limitations. It depends on region-specific cost of capital data and incentive structures, which can vary and evolve over time. This introduces subjectivity and potential volatility. Moreover, overreliance on favourable financing terms may obscure weaknesses in a project's technical or operational fundamentals. Therefore, the financing ratio should be used alongside robust techno-economic assessments to ensure a balanced and realistic evaluation of project viability (Baker Botts, 2025).

Key Insights and Drivers of the SDEI

CapEx, projected revenue, and the project-specific weighted average cost of capital ($WACC_{\text{project}}$) are the most influential variables in determining the Bankability Score, given their direct impact on discounted cash flows and overall investment viability. In parallel, energy consumption and emissions intensity are the primary drivers of the Carbon Efficiency Score (CES), as they determine the operational carbon footprint per unit of digital output. From the decarbonization perspective, the proportion of renewable energy utilized, and the scale of efficiency improvements are the most consequential inputs to the Decarbonization Readiness Score (DRS). Since SDEI integrates all three dimensions - bankability, carbon efficiency, and decarbonization readiness - these variables collectively shape a facility's composite overall sustainability profile and investment appeal.

Financing Opportunities & Bankability

Companies are increasingly seeking alternative financing strategies to support the construction and expansion of data centers, which are extremely capital-intensive. The landscape is rapidly evolving, with deal activity accelerating as developers and hyperscalers vie to build out data centers and supporting infrastructure. Even the largest, most cash-rich hyperscalers are seeking alternative financing structures due to the high costs involved ([Martin and Peskoe 2025](#)).

Data center financing can be complex, with multiple approaches to structuring deals. Companies must navigate various financing options for each part of the value chain and weigh their own benefits and challenges. Aligning financing strategies with the company's long-term vision is crucial. As data centers scale up, the capital requirements for a single data center can run into billions. As the demand for data centers grows and as the government continues to make data centers a strategic platform several financing options are becoming available.

Financing due diligence requires a broad range of subject matter experts to sign off prior to a final investment decision as several risks need to be considered and managed. A thorough operational review covering structural assessments, cooling, energy systems (data center and region), capacity, security and ESG (Environment, Social and Governance) considerations. There is also technical review of the construction covering zones, environmental plan, permits, design and power connections. Ongoing operations and maintenance plans to review operations costs and opine on technology lifecycle, as rapid advancements can render key assets/components obsolete, thus requiring ongoing upgrades for maintenance and replacement. Although blueprints for existing data centers can be utilized, a detailed capital expenditure and budget review covering construction process, drawdown schedule and timing will need to be vetted.

The contract terms for financing will range based on the different aims of the financier and alignment with the development. Contract terms include review of tenants, payment terms, change of control, exit strategy, termination, and liabilities. As the sector matures, companies must align financing strategies with stakeholder's long-term goals to navigate risks effectively.

Table 4—Data Center Financing Options, Benefits, and Challenges

Financing Option	Key Advantages	Key Considerations	Real-World Example
Project Finance	Predictable repayment; potential lower cost of capital; off-balance-sheet	Requires contracted revenue and strong credit; high due diligence	\$411M loan from DBS Bank & UOB (United Overseas Bank) to fund DayOne-INA (Indonesian Investment Authority) 72 MW data center in Batam, Indonesia (Reuters 2025a)
Corporate Financing	Low cost for investment grade issuers; flexible use of proceeds	Ties project to corporate balance sheet; depends on parent company credit rating	Equinix \$4.9B green bonds issued to fund global data center expansion (Data Frontier 2025; Gupta et al. 2025). Digital Realty Announces Pricing of €850 million of Guaranteed Notes due 2034 (Digital Realty 2025)
Private Equity & Infrastructure Funds	Provides scale and expertise; flexible JV structures	Requires IRR alignment and defined exit strategy	Meta \$29B deal: \$3B equity + \$26B private credit from Apollo, Brookfield, Carlyle for AI data centers (Global Data Center Hub 2025 ; Reuters 2025e)
REITs (Real Estate Investment Trust) & Sale-Leaseback	Tax efficient; asset monetization	Alignment on operational control	AT&T (American Telephone and Telegraph) \$850M sale-leaseback of 74 network sites with continued operational control (Reuters 2025b)
Sovereign Wealth & Pension Funds	Long-term focus, low-cost capital	Targets large, de-risked assets; slower decision cycles	\$9.2B Vantage Data Centers equity round led by DigitalBridge & AustralianSuper (Data Center Dynamics 2025a)
Green & Sustainability-Linked Loans	Access to ESG-aligned capital; meets sustainability mandates	Requires meeting ESG/KPI performance; reporting obligations	SingTel S\$643M (\$476M) green loan to develop 58 MW AI-ready sustainable data center in Singapore (Reuters 2025c)
Asset-Backed Securities (ABS)	Unlocks cheaper capital from contracted cash flows; attractive to investors	Requires long-term credit worthy tenants; structuring complexity	Vantage securitizations of lease-backed cash flows to finance expansion (Vantage Data Centers 2025)
Vendor & Equipment Financing	Reduces upfront CapEx; aligns payments with usage and cash flow	Credit risk split between facility and vendors	Applied Digital and CoreWeave ink 15-year lease worth \$7 billion (Reuters 2025d)
Public-Private Partnerships (PPP)	Subsidies, tax incentives, and land rights; aligns public interest	Strategic/regional facilities; complex stakeholder alignment	Indonesia's Ministry of National Development Planning (Bappenas) launched a public-private partnership scheme to drive digital transformation and energy transition in the country (GovInsider 2025).

Case study of profitability to test conventional vs decarbonized data centers

A case study was conducted to move beyond a simple profitability comparison and strategically evaluate conventional versus decarbonized data centers. This study uses the previously proposed Sustainable Digital Efficiency Index (SDEI) to provide a holistic assessment, analysing not just financial metrics but also carbon efficiency, decarbonization readiness, and overall bankability.

Part 1- DC sizes

Based on the collected data, it was determined that data centers worldwide can generally be classified into four categories: small, medium, large, and hyperscale data centers ([Table 5](#)).

Table 5—SDEI Input Assumptions of Data Center Scenarios

Company Size	Small	Medium	Large	Hyperscale
CapEx Base (MM\$)	0.25 - 1	1.25 - 5	5 - 20	50 - 200
CapEx_per_MW (MM\$)	3.5 - 14	4.5 - 18	5 - 20	5 - 20
Fixed OpEx (MM\$/yr)	0.5 - 2	2.5 - 10	10 - 40	100 - 400
Fixed Revenue (MM\$/yr)	1.25 - 5	6 - 24	25 - 100	250 - 1000
Revenue_per_MW (MM\$/yr)	1 - 4	2.5 - 10	4 - 16	12.5 - 50
WACC_project (%)	0.04 - 0.16	0.035 - 0.14	0.03 - 0.12	0.025 - 0.1
WACC_baseline (%)	0.05 - 0.2	0.05 - 0.2	0.05 - 0.2	0.05 - 0.2
Project Horizon (T) - Years	10 - 40	10 - 40	10 - 40	10 - 40
Discount Factor (r) - %	0.05 - 0.2	0.05 - 0.2	0.05 - 0.2	0.05 - 0.2
Capacity (Power_MW) (MW)	0.25 - 1	2.5 - 10	12.5 - 50	100 - 400
Uptime (%)	0.3 - 1.2	0.3 - 1.2	0.3 - 1.2	0.3 - 1.2
Purchased_Power_Cost (\$/KWh)	0.045 - 0.18	0.045 - 0.18	0.045 - 0.18	0.045 - 0.18
Purchased_Power_Fraction (%)	0.5 - 2	0.45 - 1.8	0.4 - 1.6	0.25 - 1
Renewable_Energy_Use (%)	0.15 - 0.6	0.2 - 0.8	0.25 - 1	0.3 - 1.2
Energy_Savings (%)	0.1 - 0.4	0.125 - 0.5	0.15 - 0.6	0.175 - 0.7
GHG_Offsets (%)	0.05 - 0.2	0.075 - 0.3	0.1 - 0.4	0.125 - 0.5
CUE_Improvement (%)	0.025 - 0.1	0.05 - 0.2	0.075 - 0.3	0.1 - 0.4
Total_Energy_Consumption (GWh/yr)	0.5 - 2	2.5 - 10	10 - 40	50 - 200
Emissions_Factor (Kg CO2/KWh)	0.25 - 1	0.225 - 0.9	0.2 - 0.8	0.175 - 0.7
Emissions_cost_per_kg (\$/Kg)	5 - 20	5 - 20	5 - 20	5 - 20
Tokens_Processed (10e9)	0.5 - 2	2.5 - 10	10 - 40	50 - 200

Capital and Operating Expenditures: CapEx Base and CapEx per MW scale significantly with size, indicating larger centers not only invest more in absolute terms but also more per unit of capacity—though this plateaus at the hyperscale level. OpEx and Revenue also increase proportionally. Hyperscale centers operate at higher revenue and cost bases, reflecting economies of scale and greater throughput.

Financial Returns & Risk Metrics: WACC (project) decreases with size, from 0.16 (Small) to 0.025 (Hyperscale), reflecting lower perceived risk and better financing terms for larger players. Baseline WACC, discount rates, and project horizons are consistent, indicating similar financial modelling frameworks.

Capacity, Uptime & Efficiency: Power capacity ranges widely, from 0.25–1 MW (Small) to 100–400 MW (Hyperscale). Uptime ranges are identical (30–120%), possibly reflecting redundancy rather than literal uptime percentage. Efficiency indicators (e.g., energy savings, renewable usage, GHG offsets, and CUE improvement) all increase with company size, showing that larger firms invest more in sustainability.

Data Processing Power: Tokens processed (proxy for AI or cloud data workloads) scale from 0.5B (Small) to 200B (Hyperscale) — a 400x increase, demonstrating the immense processing advantage of hyperscale operations.

Sustainability Metrics: Renewable use, energy savings, and GHG offsets all scale up with size, showing hyperscalers are leaders in green strategies. Emissions factors and CUE (Carbon Usage Effectiveness) improvements follow the same trend.

In summary, Hyperscale firms show the best unit economics, efficiency, and sustainability, while small firms face higher risk, cost per unit, and limited scale. Mid and Large companies act as transitional tiers, with

increasing investments in sustainability and capacity. There is a clear correlation between size, financial optimization, and environmental performance.

Figure 8 shows a sensitivity analysis of various data center performance scores (SDEI, B, CES, DRS) in response to $\pm 50\%$ variation in input parameters across company sizes (likely from Small to Hyperscale).

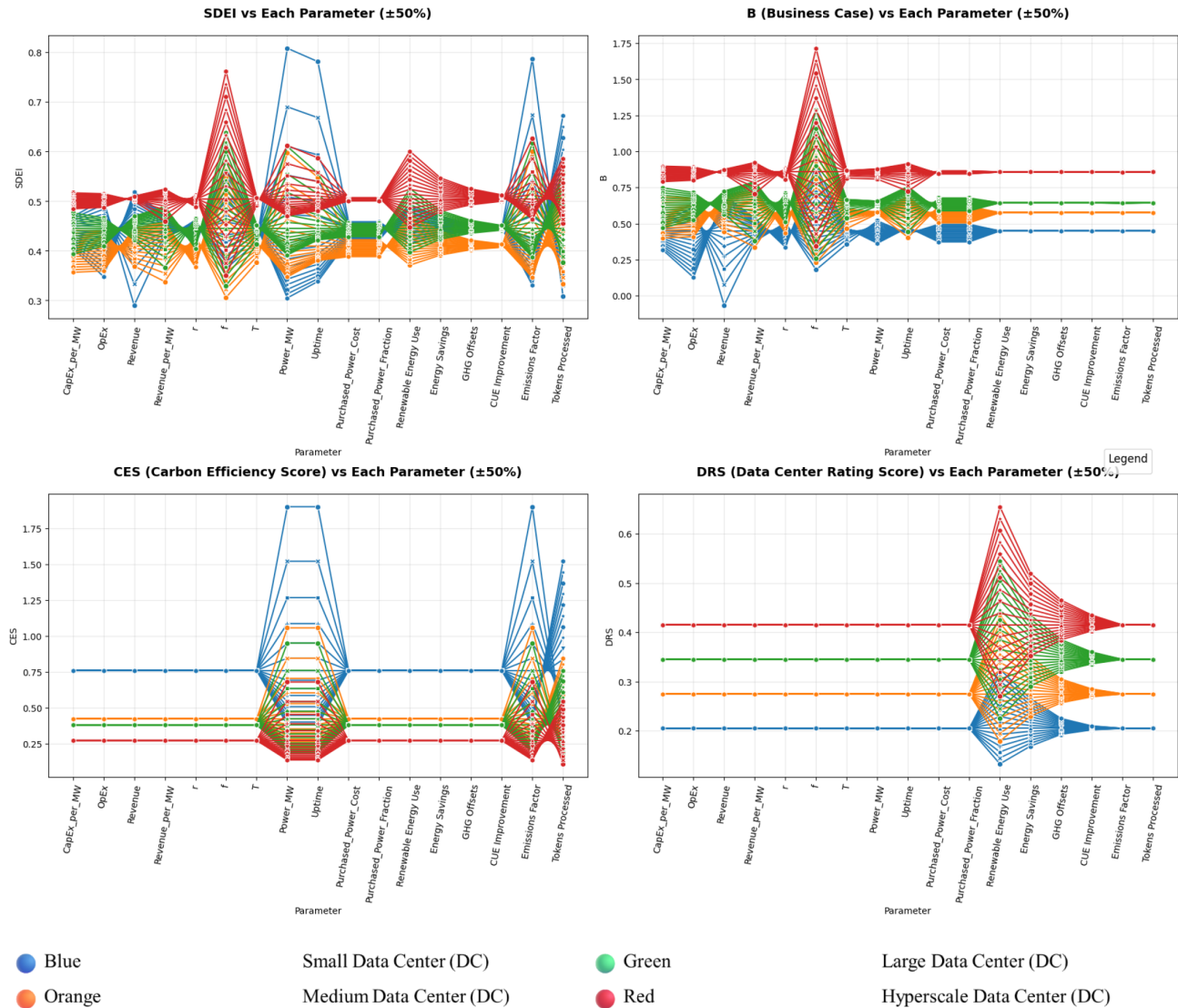


Figure 8—Sensitivity Line Plot for SDEI, Bankability, CES and DRS

SDEI (Sustainable Data Center Efficiency Index) (Top center) is most sensitive to revenue and installed power (MW), i.e. higher variation indicates these greatly affect SDEI. Cost per installed capacity (CapEx_per_MW), T (Horizon), and CUE Improvement also show noticeable impact. SDEI is relatively less affected by Emissions Cost, Tokens Processed and Purchased Power Cost with minimal influence on SDEI.

B (Bankability) (Top Right) is highly sensitive to revenue and r (Discount Rate) — large spread indicates business feasibility heavily depends on these. OpEx, CapEx_per_MW, and Power also affect results significantly. B is less affected by environmental parameters (e.g., GHG offsets, CES-related metrics), suggesting business case is more financially driven.

CES (Carbon Efficiency Score) (Bottom Center) is most sensitive to power (MW) and Uptime — carbon footprint closely tied to size and usage. CUE Improvement, Renewable Energy Use, and Energy Savings

strongly affect CES — as expected. There is minimal influence from financials (CapEx, Revenue, OpEx), r , and T , since CES is an environmental metric.

DRS (Data Center Rating Score) (Bottom Right) is strongly sensitive to renewable energy use, energy savings, and CUE Improvement — confirming that sustainability is central to this score. Purchased power fraction also plays a notable role. DRS is relatively insensitive to CapEx, OpEx, Revenue, Tokens Processed, etc. — confirming DRS is not profit-driven.

Figure 9 shows parameter-level sensitivity of the SDEI for each company size: Small, Medium, Large, and Hyperscale. It provides marginal change impact on SDEI when each parameter is varied $\pm 50\%$. The key insights are provided below.

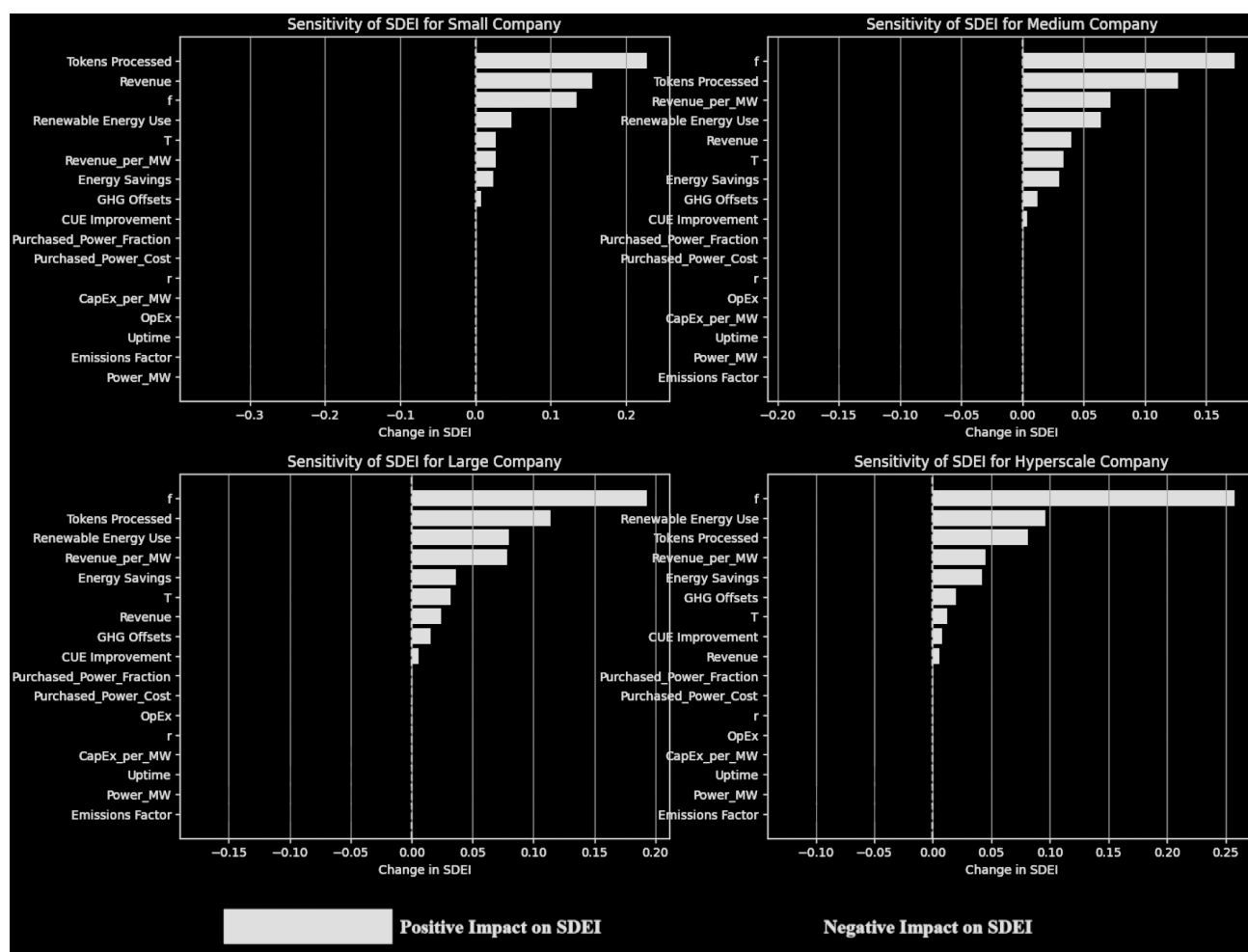


Figure 9—SDEI Sensitivities to Input Parameters for Four Data Center Categories

Positive Drivers of SDEI Vary with Scale: Small & Medium companies benefit most from Revenue, Tokens Processed, and Renewable Energy Use — suggesting a strong dependence on financial inflow and throughput. Hyperscale companies are uniquely sensitive to Renewable Energy Use as the dominant positive driver — possibly due to the larger base power consumption making clean energy more impactful. Tokens Processed also plays a strong role, emphasizing compute efficiency at scale.

Key Negative Drivers Shift with Size: Emissions Factor and Power (MW) are consistently the most negative influencers across all sizes. Reflects that higher scale amplifies absolute energy and emissions, worsening SDEI unless mitigated. OpEx and CapEx per MW have a smaller negative impact for Hyperscale compared to Small firms — indicating better economies of scale.

Uptime Becomes Less Risky at Scale: For Small firms, uptime has a strong negative effect (likely due to inefficient energy overhead). For Hyperscale, it becomes less sensitive, suggesting better resource balancing and redundancy.

The Role of Financing Factor, F: $F = \text{WACC_baseline} / \text{WACC_project}$ shows increasing positive impact as company size grows. At Hyperscale, it's the most impactful single parameter (+0.25 SDEI), highlighting the importance of long-term project valuation or ROI frameworks in large-scale projects. F has a consistently strong and positive impact on SDEI for all company sizes. The effect increases with size. For Small Companies, F is impactful but secondary to Revenue and Tokens Processed. For Hyperscale: F is the top driver of SDEI, indicating it's critical for large firms to secure low-risk, low-cost capital (i.e., low WACC_project vs WACC_baseline).

Implication for Strategy: Smaller companies should prioritize improving revenue generation and reducing scale-related inefficiencies. Larger and Hyperscale firms should focus on maximizing renewable energy adoption, minimizing emissions intensity, and optimizing long-term financial levers. For Hyperscale firms, lowering project-specific WACC (e.g., through green financing, public-private partnerships, or sovereign backing) supercharges SDEI, even more than operational or environmental efficiency improvements. For Smaller firms, improvements in F still help, but revenue generation and throughput (Tokens Processed) carry more weight—likely due to smaller project scale and tighter margins.

Part 2- Location as a driver –UAE Onshore/Offshore (SDEI)

Based on the technical analysis- see the GCC map in the [Figure 6](#), the UAE represent a good location for the DCs based on its infrastructure, regulations and ambitious plans in this domain. Location is a key driver to be analysed in the UAE, two cases were selected onshore and offshore locations. Onshore UAE assumes a land-based facility connected to the national grid, leveraging the UAE's strong transmission infrastructure, Barakah nuclear capacity, and large-scale solar PPAs. Cooling relies on advanced air-to-liquid systems thermal envelopes, with PUE typically >1.6 under hot-climate conditions. Offshore UAE represents a maritime repurposing model using end-of-life offshore assets or barges. This scenario benefits from direct seawater cooling (enabling PUE <1.2), maximized WUE, and circular-economy gains by avoiding new-build emissions. Power is supplied via subsea cable from the UAE's low-carbon grid, with optional hybridization using renewables, see [table 6](#) and [7](#).

Table 6—SDEI Comparative Evaluation of Data Center Scenarios

SDEI Component	Scenario 1: Onshore UAE	Scenario 2: Offshore UAE (High Potential)
Carbon Efficiency (CES) Score	Medium. Inherently inefficient due to reliance on energy-intensive air cooling (PUE >1.6), resulting in high carbon intensity per token processed unless connected to renewable grid.	High. Achieves high efficiency (PUE <1.2) via direct seawater cooling and connection to renewable grid. The core technology drives a low carbon intensity per token.
Decarbonization Readiness (DRS) Score	High. Limited decarbonization levers beyond purchasing grid-scale renewables. Lacks access to seawater cooling and circular economy benefits.	Very High. Maximizes readiness by combining circular economy (repurposing), direct access to seawater, and grid connection to the UAE's massive solar and nuclear capacity.
Bankability (B) Score	High. High CapEx (land), long deployment timeline (#5 years) delays revenue, and high OpEx risk from volatile energy costs make it the least attractive financial model.	Very High. Lowers CapEx (no land), accelerates revenue (#18 months), and de-risks OpEx. Perfectly aligns with UAE national strategies, attracting sovereign wealth and green financing.

The offshore scenario proposes using a shore-tied subsea cable to connect offshore operations to the UAE's low-carbon grid, which includes nuclear and renewable energy sources. This model reflects the existing option of Onshore Power to Offshore Assets approach, proving its technical and regulatory feasibility. Capital expenditure is minimized by repurposing existing maritime assets scheduled for disposal like jack-ups or barges, avoiding the need for costly new platforms or land acquisition. The study explicitly

excludes scenarios involving new-build platforms, as they would significantly alter the economic viability and fall outside the scope of this analysis.

Table 7—SDEI Score Calculation for Data Center Scenarios

SDEI Component	Weighting	Scenario 1: Onshore UAE (Score)	Scenario 2: Offshore UAE (Score)
Carbon Efficiency (CES)	30%	6	8
<i>Weighted Score</i>		<i>1.8</i>	<i>2.4</i>
Decarbonization Readiness (DRS)	30%	7	8
<i>Weighted Score</i>		<i>2.1</i>	<i>2.4</i>
Bankability (B)	40%	7	8
<i>Weighted Score</i>		<i>2.8</i>	<i>3.2</i>
Final SDEI Score		6.7	8.0
Qualitative Ranking		Lagging	Advanced

In summary, repurposing maritime assets as offshore data centers may have significant potential. This approach bypasses the need for extracting/manufacturing new, emission-intensive materials like concrete and steel by reusing assets. It also prevents the environmental impact and energy consumption of scrapping the vessel, creating a net positive from a circular economy standpoint. For operations, the design leverages direct access to the Arabian Gulf's deep seawater, enabling highly efficient cooling systems that drastically lower the PUE, a critical factor in hot climates.

The financial structure of this model is equally innovative. A drastic reduction in initial CapEx is the primary benefit, as acquiring an end-of-life vessel is substantially cheaper than land acquisition and new construction. The project effectively transforms a vessel owner's decommissioning liability into an asset, creating a uniquely favorable acquisition scenario. This capital efficiency, combined with a faster time-to-market and a modular IT fit-out, directly improves key financial metrics like the IRR. As a result, the project's risk profile drops, lowering its Weighted Average Cost of Capital (WACC) and making it a compelling proposition for investors.

These technical and financial advantages are magnified within the UAE. The project's decarbonization potential is amplified by direct integration with the UAE's robust and rapidly renewable grid, which includes world-class solar projects and the Barakah nuclear plant. Crucially, its bankability is secured by its perfect alignment with national strategic goals, including the UAE Net Zero by 2050 initiative. This alignment unlocks access to preferential financing from sovereign wealth funds and supports the regulatory pathway. This unique combination of innovative design, financial logic, and strategic national support can establish a new global benchmark for developing profitable and environmentally responsible digital infrastructure. This allows companies to create a replicable and scalable model for future digital infrastructure that is both profitable and environmentally responsible.

UAE Governance

Among the governance models shaping the UAE's data center landscape, the Stargate consortium stands out as the most comprehensive and scalable. This initiative, led by G42 ([G42 2025](#)) in collaboration with OpenAI, Oracle, Cisco, and NVIDIA, operates under a multi-stakeholder governance framework that balances sovereign oversight with international expertise. Each partner assumes specialized roles—G42 leads infrastructure, Oracle and OpenAI manage AI operations, and Cisco oversees cybersecurity—while regulatory compliance is maintained with both UAE's Personal Data Protection Law (PDPL) and US export

controls ([Sharwood 2025](#); [Cisco 2025](#)). This structure enables ethical oversight, geopolitical accountability, and operational agility, making it particularly suited for hyperscale AI infrastructure.

In contrast, Khazna Data Centers, majority-owned by G42 and e& (Etisalat), follows a centralized governance model aligned with national digital strategies. Governed by the Telecommunications and Digital Government Regulatory Authority (TDRA) and the Information Assurance Regulation (IAR), Khazna emphasizes data sovereignty, cybersecurity, and regulatory compliance ([Khazna 2025](#); [Eni 2025](#)). While this model ensures strong alignment with UAE policy and risk management protocols, it lacks the international integration and distributed operational flexibility seen in Stargate's framework.

Meanwhile, the Kinetics-Mitsui model adopts a non-equity-based governance approach defined by a Memorandum of Understanding (MOU), focusing on technical feasibility and regulatory navigation without a formal board structure. This agile model supports innovation in floating data centers powered by powerships and renewables, with capacities ranging from 20 to 73 MW ([Mitsui O.S.K. Lines 2025](#); [Data Center Dynamics 2025b](#)). However, its lack of institutional depth and formal governance mechanisms makes it less suitable for large-scale, sovereign-aligned AI infrastructure. Overall, Stargate's consortium model offers the most robust governance framework, integrating global partnerships, regulatory rigor, and operational specialization.

For the case study, the authors recommend a deep-dive evaluation to assess the best financing options, depending on the company's preferences and identified risks.

Conclusions

Decarbonizing data centers in the GCC presents both challenges and opportunities shaped by regional conditions, regulatory evolution, and technological maturity. Key challenges include high initial capital costs for advanced cooling and renewable integration, climatic constraints such as extreme heat and water scarcity that complicate energy and water efficiency, and regulatory uncertainty due to evolving compliance frameworks. Additionally, emerging technologies like subsea data centers and offshore repurposing are still in early stages, requiring careful risk assessment, while rapid AI-driven load growth may strain grid flexibility.

The rapid expansion of digital infrastructure, driven by AI and cloud adoption, places data centers at the center of the energy transition challenge. For the GCC, this challenge is amplified by extreme climatic conditions but offset by unique advantages: a rapidly decarbonizing power mix, ambitious national strategies, and access to sovereign capital. Growing momentum in green financing and supportive national policies such as UAE Net Zero 2050 and KSA Vision 2030 further enhance the region's potential as a testbed for scalable innovations in sustainable digital infrastructure.

This paper introduces SDEI as an integrated framework to evaluate carbon efficiency, decarbonization readiness, and bankability—three dimensions critical for aligning technical performance with financial viability. Our comparative analysis of onshore and offshore scenarios demonstrates that innovative approaches, such as maritime asset repurposing combined with low-carbon grid access, can outperform conventional models on both sustainability and cost metrics.

Future work aims to focus on refining the SDEI methodology, incorporating empirical financing data through lender surveys, and validating offshore and subsea concepts under real-world conditions. By leveraging regional strengths and addressing identified challenges, the GCC can position itself as a global benchmark for sustainable, high-performance digital infrastructure.

Acknowledgments

We thank all individuals who provided their expertise and support to this study. This research was made possible by the use of open-source data from various organizations and articles. We also acknowledge the assistance of AI language models for improving the clarity and readability of this manuscript.

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